MADHU MOWLI

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PROFILE

Applied Computing (FinTech) undergraduate with a strong interest in **software engineering, AI-powered enterprise systems, and large-scale digital workflows**. Experienced in building, validating, and maintaining **data-driven backend workflows**, system integration, and automation logic using Python, C, and SQL fundamentals. Hands-on experience integrating **LLM-powered features** into web applications, including AI chatbot development using prompt engineering. Comfortable working on **real-world systems** and building **reliable, well-documented software** that supports operational and user-facing services.

PROFESSIONAL EXPERIENCE

Student Intern, Sick Pte Ltd **Mar 2022 to Aug 2022**
Singapore

- Analysed **operational and system-level metrics** to identify reliability issues and support platform improvements in a **production environment**.
- Built and maintained **data processing and monitoring workflows** that consolidated inputs from multiple system sources.
- Designed **logic-based automation pipelines** to reduce manual intervention and improve data consistency across systems.
- Supported **application deployment, performance tracking, and issue resolution**, working closely with engineers during live operations.
- Collaborated with cross-functional stakeholders to **validate changes, interpret system behaviour, and improve workflows**.

EDUCATION

Bachelor's of Science in Applied Computing (Fintech) **Sept 2025 to Sept 2028**
Singapore Institute of Technology, Singapore

CGPA: 4.5 / 5.0 (Year 1, Trimester 1)

Relevant Coursework & Skills:

- Programming Fundamentals (**Python, C**)
- Object-Oriented Programming (**Java**)
- Web Systems & Technologies (**HTML, CSS, JavaScript**)
- Data Structures & Algorithms
- RStudio (**Data Visualisations**)

AREAS OF EXPERTISE

Data Processing & Validation · Process Automation & Workflow Logic · Financial & Operational Systems (ERP-style workflows) · Programming: Python, C, Java, JavaScript · Data Analysis & Visualisation (RStudio, Quarto, Excel-equivalent logic) · Tools: Git, Linux, SQL, REST APIs · Prompt Engineering · LLM Integration · AI Chatbot Development · HTTP, DNS & Web Fundamentals · Testing, Documentation & Process Control

PROJECT EXPERIENCE

Web Systems and Technologies **Mar 2026 to Apr 2026**
Singapore Institute of Technology

- Designed and developed a full-stack e-commerce web application using **PHP**, **MySQL**, **Bootstrap**, and **JavaScript**, **hosted** on **Google Cloud Platform**.
- Integrated an **AI-powered chatbot** using **Claude's LLM API,** designing structured prompts and handling conversation flows to assist users with product queries.
- Built **REST API** endpoints for dynamic data retrieval, including store locator integration with **Google Maps API**.

Data Engineering & Visualisation (R programming) **Feb 2026 to Mar 2026**
Singapore Institute of Technology

- Analysed datasets using R to identify trends and generate insights.
- Cleaned, transformed, and aggregated data using **tidyverse** workflows.
- Built clear visualisations and dashboards using **ggplot2** and **Quarto**.

Object-Oriented Programing Abstract & Game Engine (Java) **Jan 2026 to Mar 2026**
Singapore Institute of Technology

- Developed a 2D game engine in **Java,** applying **Factory, Strategy, and Observer** design patterns to decouple engine and game logic.
- Built a **Scene/SceneManager** system for managing game states and transitions.
- Refactored codebase into a structured package hierarchy with full **Javadoc** documentation.

Student CMS (C Programming) **Nov 2025 to Dec 2025**
Singapore Institute of Technology, Grade A

- Designed and implemented a structured backend system supporting full **CRUD operations**.
- Applied **input validation, memory management, and data integrity checks** to ensure system stability.
- Developed clear control flows for **data processing, error handling, and system workflows**.

Stock Market Analysis System (Python) **Sept 2025 to Oct 2025**
Singapore Institute of Technology, Grade A

- Built a **Python-based data analysis system** to process and analyse financial market data.
- Performed **data cleaning, exploratory data analysis (EDA), and computational analysis** to identify trends and patterns.
- Developed **data visualisations** to communicate insights clearly and support analytical conclusions.
- Applied version control using **Git** to ensure reproducibility and maintainability of analysis workflows.

CERTIFICATIONS

Fundamentals of Deep Learning , NVIDIA	Feb 2026 to Feb 2026
Blockchain Technology and Web3 Projects , Hooked / Defi App / SIT	Oct 2025 to Oct 2025
The Bits and Bytes of Computer Networking, Coursera	Jan 2025 to Feb 2025
C for Everyone: Programming Fundamentals, Coursera	Dec 2024 to Feb 2025
Blockchain Basics, Coursera	Feb 2024 to Feb 2024
Internet of Things, Singapore Polytechnic	Jan 2020 to Dec 2023